

Index in SQL

SQL Indexes is used in relational databases to quickly retrieve data. They are similar to indexes at the end of the books whose purpose is to find a topic quickly. SQL provides Create Index, Alter Index, and Drop Index commands that are used to create a new index, update an existing index, and delete an index in SQL Server.

Type of Indexes:

SQL Server supports two types of indexes:

1. Clustered Index
2. Non-Clustered Index.

1. **Clustered Index:** A B-Tree (computed) clustered index is the index that will arrange the rows physically in the memory in sorted order. When the Primary Key is created automatically the Clustered **Index is created**.

An advantage of a clustered index is that searching for a range of values will be fast. A clustered index is internally maintained using a B-Tree data structure leaf node of the b tree of clustered indexes that will contain the table data; **you can create only one clustered index for a table**.

2. Non-clustered Index:

- When a Unique Key is created automatically, a Non-clustered Index is created.
- A non-clustered index is an index that will not arrange the rows physically in the memory in sorted order.
- An advantage of a non-clustered index is searching for the values that are in a range will be fast.
- You can create a maximum of 999 non-clustered indexes on a table, which is 254 up to SQL Server 2005.
- A non-clustered index is also maintained in a B-Tree data structure but leaf nodes of a B-Tree or non-clustered index contain the pointers to the pages that contain the table data and not the table data directly.

Clustered Index	Non- Clustered Index
This will arrange the rows physically in the memory in sorted order	This will not arrange the rows physically in the memory in sorted order.
This will fast in searching for the range of values.	This will be fast in searching for the values that are not in the range.
Index for table.	You can create a maximum of 999 non clustered indexes for tables.
Leaf node of the 3 tier of clustered index contains table data.	Leaf nodes of b-tree of non-clustered index contain pointers to get the contained pointers that contain two table data, and not the table data directly.